

Reviewer Pack — Minimal Local Cohomology Rank of Quiver Representations vs R...

Entry 7cca45028040 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Local Cohomology Rank of Quiver Representations vs Resolution Proof Length for Non-clausal Tseitin Formulas

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1. Conjecture

Field A (mathematical branch): Algebraic Combinatorics (Quiver Representations) **Field B** (complexity object): Complexity Theory: Resolution Proof Complexity for Non-clausal Tseitin Formulas

Statement:

[‘For a given non-clausal Tseitin formula φ on n variables, the length of its shortest resolution proof is at least $2^{\Omega(h(\varphi))}$, where $h(\varphi)$ is the minimal local cohomology rank of any quiver representation associated with φ .’, ‘Further, for all non-clausal Tseitin formulas φ , $h(\varphi) = O(\log n)$.’, ‘The local cohomology rank $h(\varphi)$ can be computed in polynomial time.’]

Rationale (proposer’s reasoning):

[‘Quiver representations encode combinatorial structures that may capture the complexity of proof systems, particularly resolution. By studying the minimal local cohomology rank of quiver representations associated with Tseitin formulas, we might uncover a new perspective on resolution complexity.’, ‘Local cohomology can be thought of as a generalization of the concept of vanishing dimensions in algebraic geometry, which has been used to study other complexity-theoretic problems.’, ‘If this conjecture holds, it would provide a stronger connection between algebraic combinatorics and proof complexity, potentially leading to new algorithms or lower bounds.’]

Taxonomy category: TSEITIN_RES (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 024a4374344b7c9e

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

For a non-clausal Tseitin formula, if the shortest resolution proof length is $\geq 2^{(\text{computed } h(\varphi))}$ for all seeds and $\text{computed } h(\varphi) \leq \log n$, then support is provided; otherwise, the conjecture is falsified.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	1.00	SAFE	UNCERTAIN
NATURAL_PROOFS	UNCERTAIN	1.00	UNCERTAIN	HITS
ALGEBRIZATION	SAFE	1.00	SAFE	SAFE
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 9 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - Minimal local cohomology rank AND quiver representations AND resolution proof length - Tseitin formulas AND non-clausal AND minimal local cohomology rank AND resolution complexity - Cohomology rank $h(\phi)$ AND polynomial time computation IN Algebraic Combinatorics AND Resolution Proof Complexity Theory

Top relevant hits considered: - [<http://arxiv.org/abs/1708.00203v2>] Hochschild cohomology of algebras arising from categories and from bounded quivers - [<http://arxiv.org/abs/math/0406355v1>] p-torsion elements in local cohomology modules. II - [<http://arxiv.org/abs/2102.04369v2>] Mixed Hodge structure on local cohomology with support in determinantal varieties - [<http://arxiv.org/abs/2407.17951v1>] Pruning Boolean d-DNNF Circuits Through Tseitin-Awareness - [<http://arxiv.org/abs/2209.05839v3>] On bounded depth proofs for Tseitin formulas on the grid; revisited - [<http://arxiv.org/abs/2103.09609v1>] Characterizing Tseitin-formulas with short regular resolution refutations - [<http://arxiv.org/abs/2604.05701v1>] Measurement of the CKM angle γ in $B^\pm \rightarrow D(\rightarrow K_S^0 h'^+ h'^-) h^\pm$ dec - [<http://arxiv.org/abs/2410.02421v2>] Search for lepton number violating decays of $D_s^+ \rightarrow h^- h^0 e^+ e^+$

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_tseitin_formula(n):
        variables = [f'x{i}' for i in range(1, n+1)]
        clauses = []
        for i in range(1, n+1):
            clause = f'{variables[i-1]}'
            clauses.append(clause)
        return variables, clauses

    def construct_quiver_representation(variables, clauses):
        n = len(variables)
        quiver = [[0] * (n + len(clauses)) for _ in range(n + len(clauses))]
        for i in range(n):
```

```

        quiver[i][i] = 1
    for clause in clauses:
        ci = variables.index(clause) + n
        for var in clause.split():
            ai = variables.index(var)
            quiver[n+ai][ci] = 1
            quiver[ci][n+ai] = 1
    return quiver

def gaussian_elimination(A):
    m, n = len(A), len(A[0])
    for i in range(m):
        max_row = i
        for j in range(i+1, m):
            if abs(A[j][i]) > abs(A[max_row][i]):
                max_row = j
        A[i], A[max_row] = A[max_row], A[i]
        if A[i][i] == 0:
            return None
        for j in range(n):
            A[i][j] /= A[i][i]
        for k in range(m):
            if k != i and A[k][i] != 0:
                factor = A[k][i]
                for j in range(n):
                    A[k][j] -= factor * A[i][j]
    return A

def compute_local_cohomology_rank(quiver):
    m, n = len(quiver), len(quiver[0])
    rank = 0
    for i in range(m):
        if quiver[i][i] != 1:
            continue
        rank += 1
        for j in range(n):
            if j != i and quiver[j][i] == 1:
                for k in range(n):
                    quiver[j][k] -= quiver[i][k]
    return rank

def resolution_proof_length(variables, clauses):
    proof_length = len(clauses)
    for clause in clauses:
        proof_length += len(clause.split())
    return proof_length

variables, clauses = generate_tseitin_formula(10) # Start with n=10
quiver_representation = construct_quiver_representation(variables, clauses)
if quiver_representation is None:
    return {
        "metric_name": "Local Cohomology Rank",
        "metric_value": float('inf'),
        "instances_tested": 1,
        "conjecture_holds": False,
        "counterexample": "mapping_undefined"
    }

```

```

    }

    local_cohomology_rank = compute_local_cohomology_rank(quiver_representation)
    proof_length = resolution_proof_length(variables, clauses)

    return {
        "metric_name": "Local Cohomology Rank",
        "metric_value": local_cohomology_rank,
        "instances_tested": 1,
        "conjecture_holds": proof_length >= 2**local_cohomology_rank and local_cohomology_rank <=
        "counterexample": ""
    }

if __name__ == "__main__":
    seeds = [int(x) for x in sys.argv[1:]] or [random.randint(2, 10**9) for _ in range(30)]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    std_value = math.sqrt(sum((r["metric_value"] - mean_value)**2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    else:
        first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
        counterexample = "mapping_undefined"
        print(f"RESULT: FALSIFIED counterexample=\"{counterexample}\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

counterexample': ''}
TRIAL: {'metric_name': 'Local Cohomology Rank', 'metric_value': 20, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'Local Cohomology Rank', 'metric_value': 20, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'Local Cohomology Rank', 'metric_value': 20, 'instances_tested': 1, 'conjecture_holds': True, 'counterexample': ''}
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--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_8498eb2d.py", line 120, in <module>

```

